

HW9 Con.

$A^{-1} \rightarrow$ solve: $Ax = e_j$
 $\uparrow$ 3 systems $\rightarrow$ one for each vector

C16. $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{bmatrix}$

1. $Ax = e_1 = (1, 0, 0)^T$
 $x = (x, y, z)^T$

$$\begin{cases} x+z=1 \rightarrow x=1-z \\ x+y+z=0 \rightarrow 1-z+y+z=0 \Rightarrow 1+y=0, y=-1 \\ 2x-y+z=0 \rightarrow 2(1-z)-(-1)+z=0 \end{cases}$$

$$\hookrightarrow 2-2z+1+z=0$$

$$\hookrightarrow 2-2z+1=0$$

Solution: $x = (-2, -1, 3)^T \rightarrow$ First column of A^{-1} $z=3$

2. $Ax = e_2 = (0, 1, 0)^T$ $x=1$

$$\begin{cases} x+z=0 \rightarrow x=-z \\ x+y+z=1 \rightarrow -z+y+z=1 \Rightarrow y=1 \\ 2x-y+z=0 \rightarrow 2(-z)-1+z=0 \end{cases}$$

Solution: $x = (1, 1, -1)^T \rightarrow$ second column of A^{-1} $\hookrightarrow -2z-1+z=0$
 $\hookrightarrow z=-1$

3. $Ax = e_3 = (0, 0, 1)^T$

$$\begin{cases} x+z=0 \rightarrow x=-z \rightarrow x=1 \\ x+y+z=0 \rightarrow -z+y+z=0 \Rightarrow y=0 \\ 2x-y+z=1 \rightarrow 2(-z)-0+z=1 \end{cases}$$

Solution: $x = (1, 0, -1)^T$ $\hookrightarrow -2z+z=1 \Rightarrow z=-1$
 $\hookrightarrow$ Third column of A^{-1}

$$A^{-1} = \begin{bmatrix} -2 & 1 & 1 \\ -1 & 1 & 0 \\ 3 & -1 & -1 \end{bmatrix}$$

$e_1 =$ check column 1:

Answer check: $A \begin{bmatrix} -2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ -1 \\ 3 \end{bmatrix}$

$$= \begin{bmatrix} 1 \cdot (-2) + 0 \cdot (-1) + 1 \cdot 3 \\ 1 \cdot (-2) + 1 \cdot (-1) + 1 \cdot 3 \\ 2 \cdot (-2) + (-1) \cdot (-1) + 1 \cdot 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \checkmark \text{ correct}$$